

COUGAR COMPASS

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Southern Illinois University Edwardsville

Problem and Solution Overview:

Visually impaired students and faculty encounter a significant challenge navigating large areas, such as the campus of SIUE. Existing applications include NaviLens and VoiceVista, but they lack key accessibility features. They do not provide turn-by-turn audio directions, are largely beacon-based at only primary locations, and do not provide any aid inside a building. QR-code-like sensors have been placed around campus, but unless a student has their phone out scanning the entire time, they are not likely to receive any information from these pieces of paper. Memorization remains a primary tool, and our goal was to develop a solution to help visually impaired individuals on campus safely and easily navigate known and new routes both indoors and outdoors, provide emergency assistance, alert users to hazards and crosswalks, and provide users with the ability to share favorite routes through a centralized database. *Cougar Compass* integrates an application for mobile phones with embedded sensors in walkways and hallways to provide real-time cues for users to reach their destination. Our system will increase users' confidence in independent mobility and include the human element of emergency services and route sharing to support users who are not as comfortable with a technological solution.

Initial Paper Prototype:

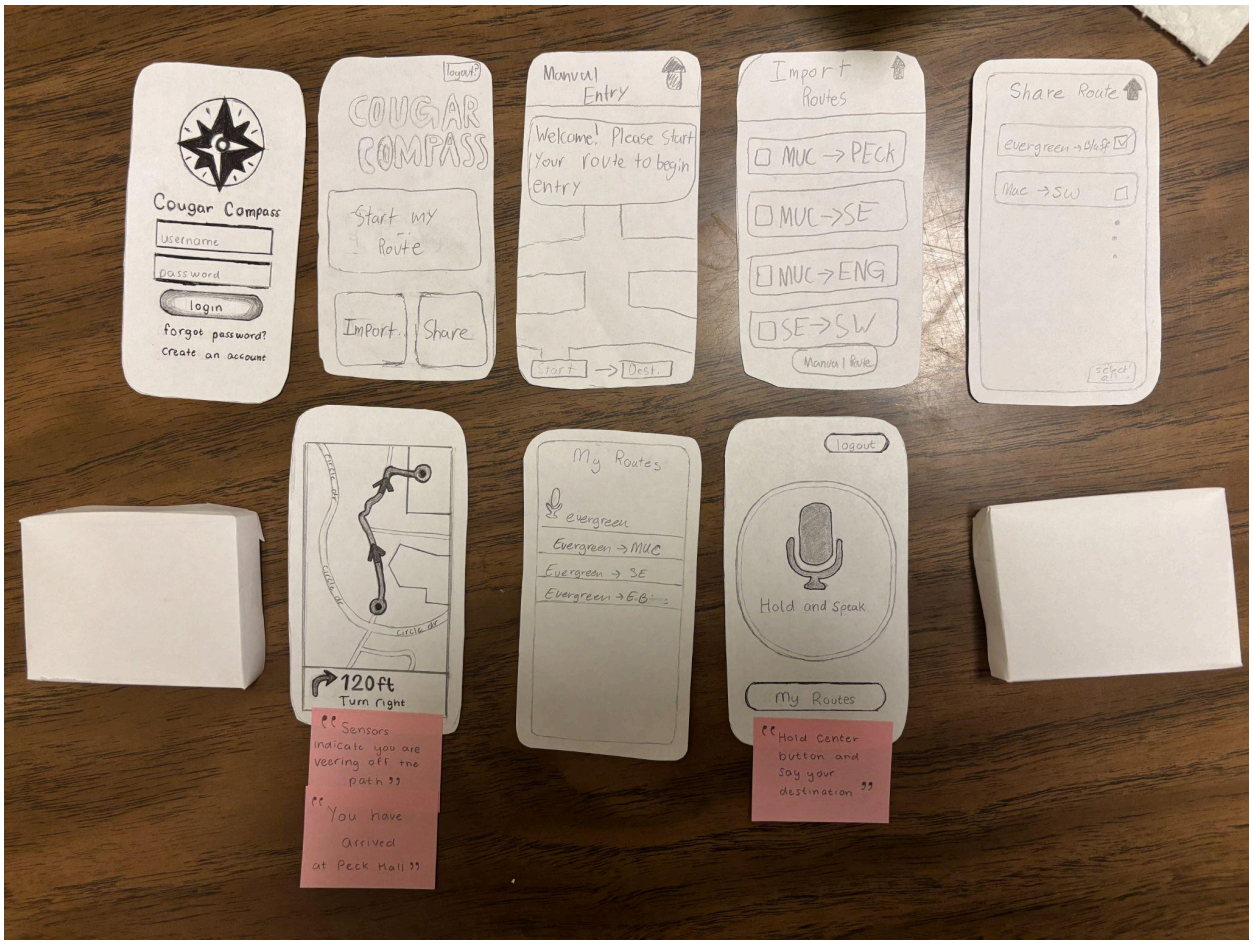
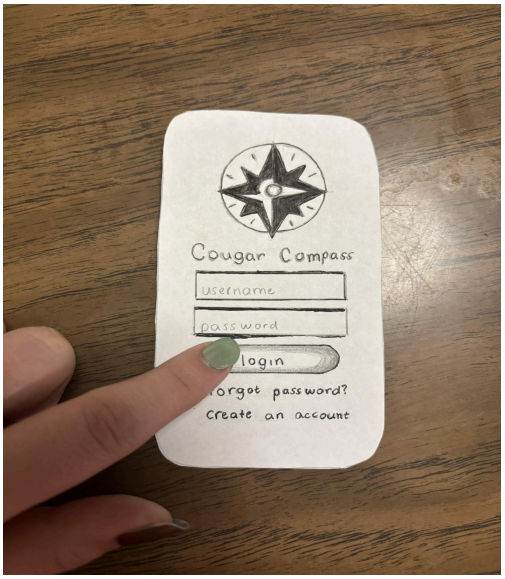


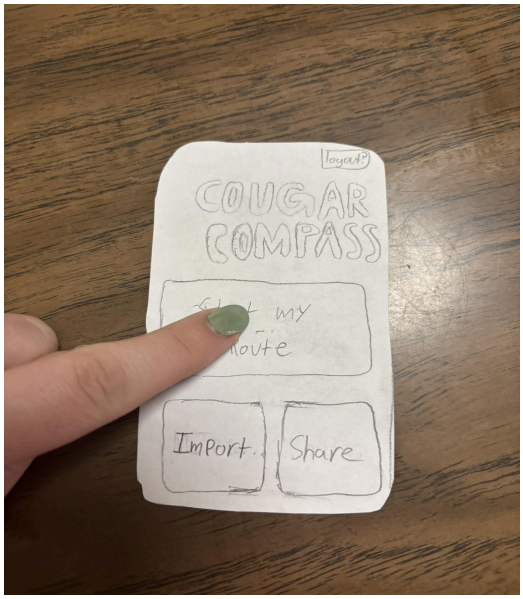
Figure 1: Initial paper prototype and embedded sensors

Cougar Compass consists of an app and a collection of embedded sensors in sidewalks and within buildings on campus. These sensors will allow for real-time audio cues as directions and keep track of a user's location should they need to contact emergency services. Pressing the side button of the phone three times will connect the user with the SIUE police. Our paper prototype focuses on two main tasks: following a route across campus and sharing existing or manually created routes with other users. Our design focused on simplicity and ease of use, as a screen reader would likely be the primary way for a user to be aware of what is on each screen. We opted for a voice-to-speech option to search for routes so users would not have to have the added step of using a special keyboard. We tried to keep similar patterns for the screens of importing and sharing for greater ease of use.

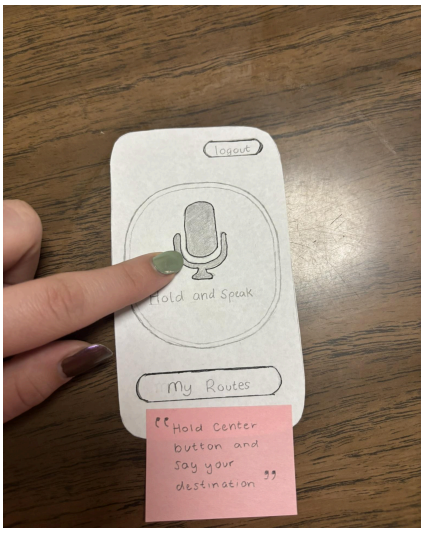
Task 1: Walking Independently Following an Existing Route



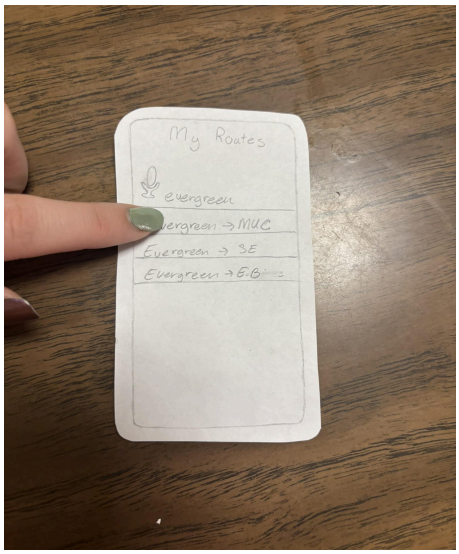
A user enters their username and password to login or creates an account.



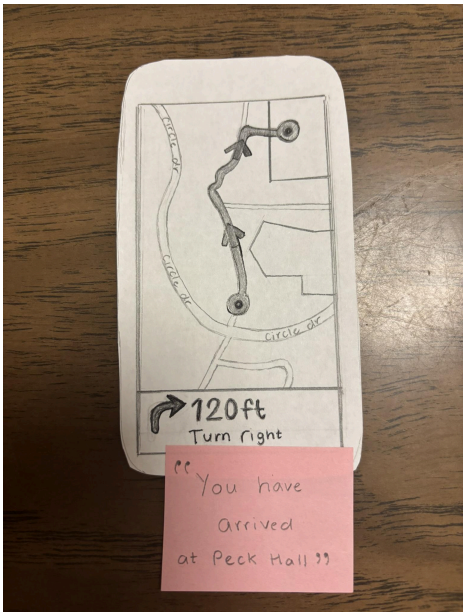
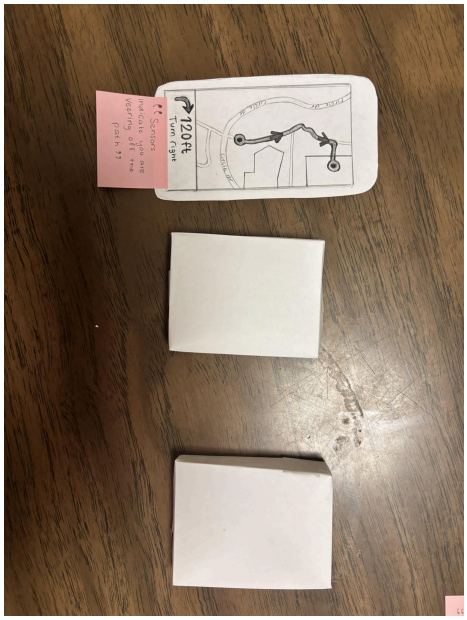
Users click “Share My Route” to search for a route.



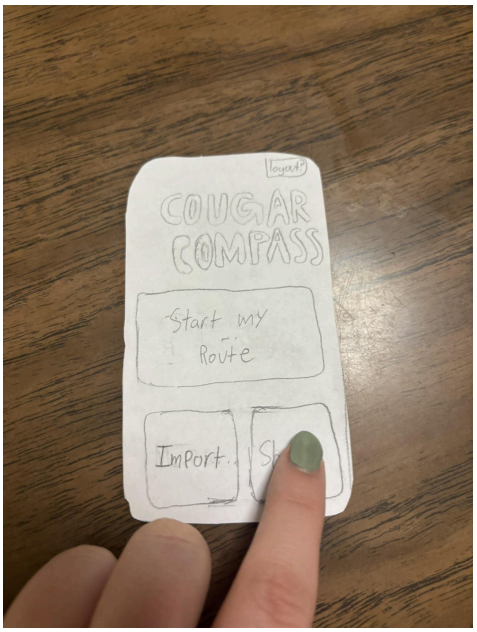
A vocal cue or screen reader will let the user know that they can click on the microphone to use voice-to-text to search for a route saved in their My Routes database.



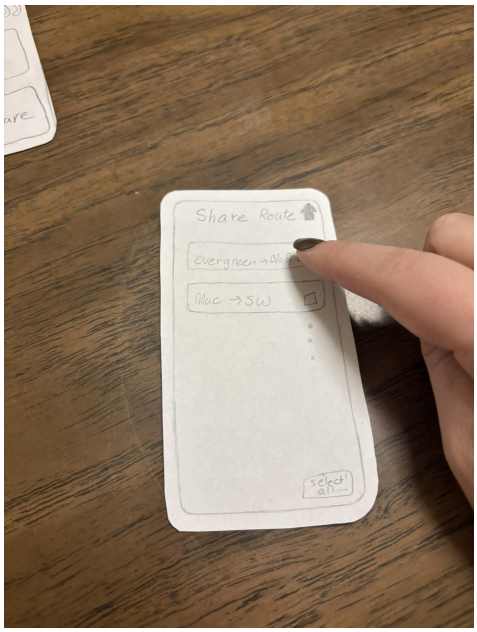
Most relevant routes appear (the user can also do a keyboard search above). The user clicks on their desired route.

	
An audio cue reads “Click once to start directions.” The user begins walking, and the app will tell them to turn left in five feet, etc. and when they have arrived.	If a user veers off the path (is not in between the sensors), the app will tell them to move more to the left or right to get back on the path.

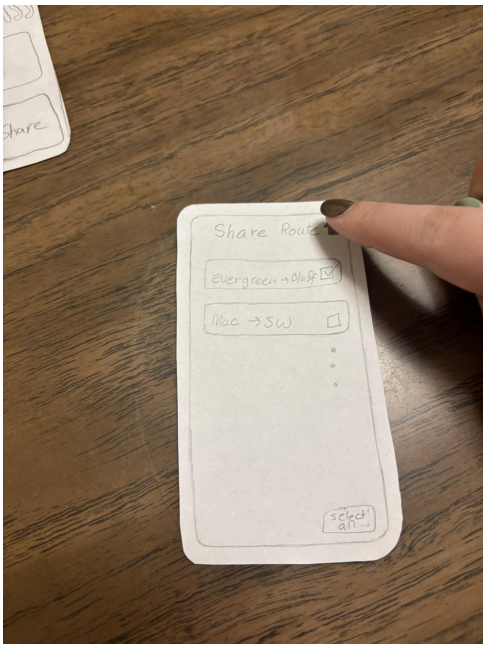
Task 2: Sharing a Route For Others (Manually Created or Existing)



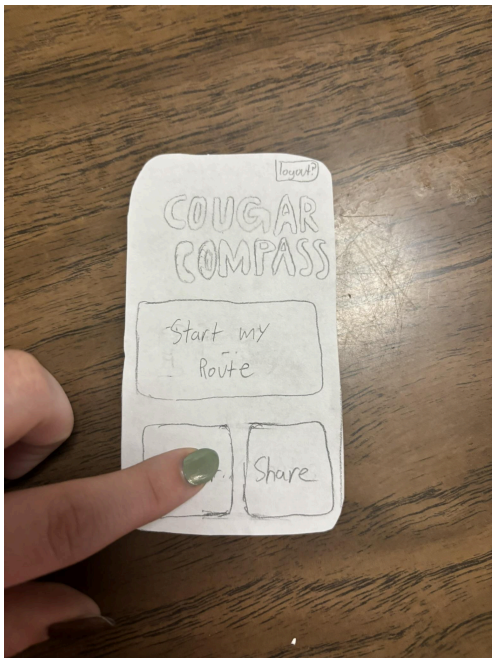
To share an existing route currently stored in the user's database, they press "Share."



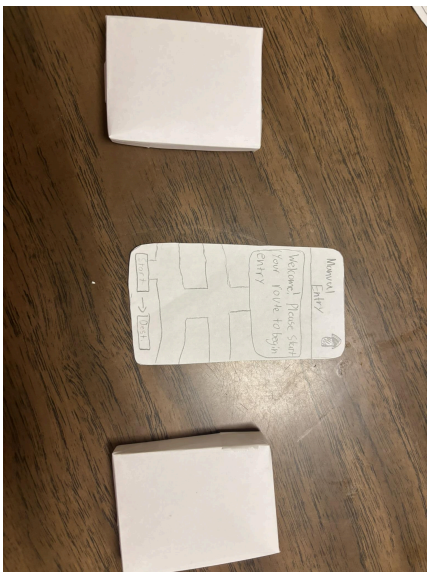
Users then select specific routes with checkboxes or "select all."



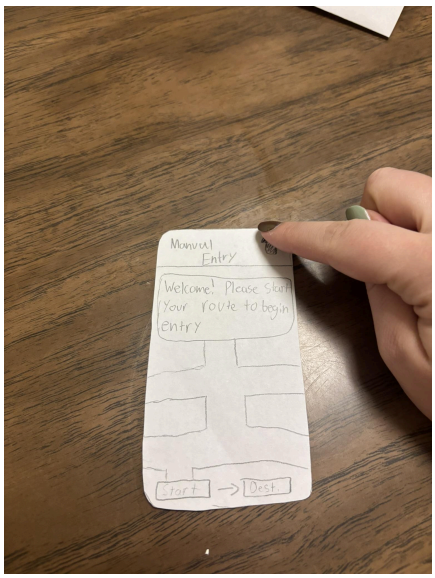
Users share selected routes to the centralized database by pressing the up arrow.



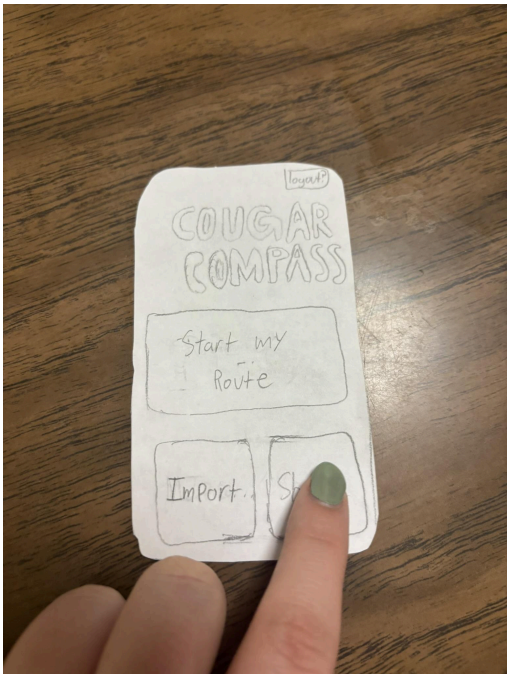
To share a new route that the student wants to manually create, they will create the route by pressing "Import."



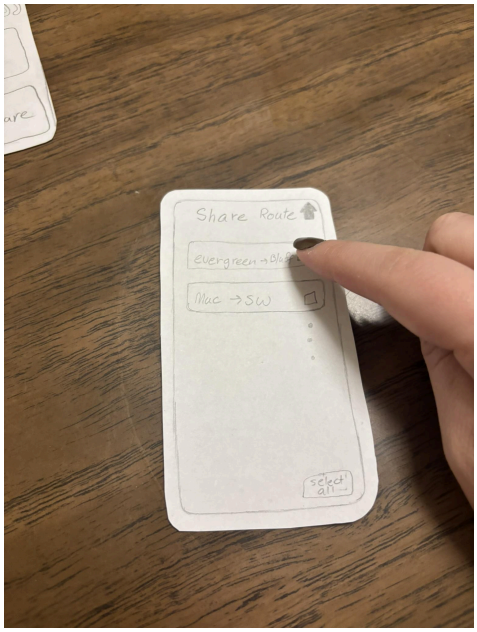
The user is prompted by an audio cue to tap the screen to begin walking. They proceed to walk the route. Embedded sensors keep track of their progress and create the route as they go.



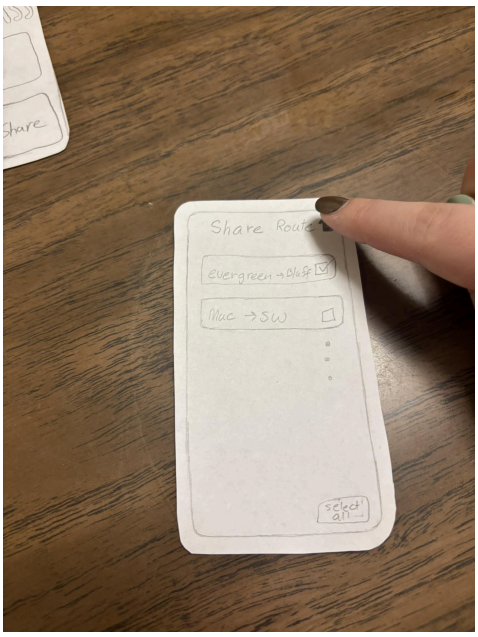
When a user has finished their manual route, they can save it to their personal “My Routes” database by pressing the up arrow in the corner.



Students are directed back to the homepage, where they press “Share.”



Users then select the created route from their own database using the checkbox.

 Users share selected routes to the centralized database by pressing the up arrow. Other users can access these shared routes using the “Import” option.	

Testing Process:

Heuristic evaluations and user testing provided critical feedback for our project. Initial heuristic evaluations were conducted with seeing members of our Human-Computer Interaction and Design (HCI) class. As these participants did not have a visual disability and had knowledge of good design principles based on HCI teachings, they provided us with valuable feedback surrounding the initial visual design and user flow, pointing out glaring errors of missing components (back buttons, create account page, etc.).

For our user testing, the following protocol was followed:

Introduction: We began by explaining that we are students in an HCI class where we were trying to create an app to help visually impaired students and faculty navigate

campus. It was explained that this project is merely the ideation stage of such a product, and a working solution likely will not come to fruition.

Tasks: We explained that the two main tasks we were testing were to navigate across campus using a new or preexisting route and to share a route to the database for others to view. We asked users to share an existing route and create a new manual route to share.

To conduct the tests, we used a paper prototype of our application. Each screen was presented one at a time to simulate the app's navigation flow. Because the prototype did not include tactile or digital accessibility features, we verbally described each screen in detail, effectively acting as a screen reader—an approach that reflects how many visually impaired users interact with mobile applications. After each description, the participant explained what actions they would take, which allowed us to evaluate whether the interface and navigation flow were intuitive. As our participants have a disability, there were a few instances where they did require a bit more prompting to complete a task, with the group at times just walking through a step and saying out loud what was happening. The provided feedback was still helpful despite this, with participants being able to identify if something seemed cluttered or had too many steps.

During testing, each team member had a defined role during the session. Aiden presented the paper prototype by placing each screen in front of the participant, Emmorie described the interface and interactions for each screen, and Faith served as the note-taker, documenting the responses, behaviors, and feedback.

Participants:

Participant 1: Our first usability test was conducted with the Blind Services Coordinator from the campus ACCESS office who regularly uses mobile applications to navigate campus. We selected this participant because of her lived experience with visual impairment and assistive technologies, making her an especially valuable participant for providing realistic and actionable feedback. The ACCESS office was chosen as the testing environment because it is a familiar, accessible space, allowing for a comfortable and natural testing experience.

Participant 2: Our second participant was a freshman student who uses a guide dog to help navigate campus. She is a relatively new student who may need more help than more experienced students navigating campus and her experiences with a guide dog make her input unique. We conducted the test just outside of the ACCESS office in the Student Success Center, allowing for a familiar testing environment.

Participant 3: Our third participant was a graduate student who will be receiving her degree at the end of the summer term. She attended a school for the blind before her time at SIUE to ease the transitional process. She is experienced at learning to navigate new places, so she was able to provide feedback concerning features she wishes she would have had access to. We conducted the test just outside of the ACCESS office in the Student Success Center, allowing for a familiar testing environment.

Our testing process changed greatly from the heuristic evaluations to the tests conducted with visually impaired individuals. The students who helped us complete the heuristic evaluation required much less prompting and we often forgot to be thorough screen readers for them. Accordingly, they were able to give us better feedback about the visual layout and structure of the app, but visually impaired users gave us feedback that seeing individuals simply could not have thought of. For tests conducted with visually impaired students, we learned to improve our verbal descriptions and standardize the cues “tap once to start, tap again to stop.” Visually impaired users struggled with the lack of tactile cues of the paper prototype, so we asked them to be very vocal about their thought process and simply tell us which button they would press next.

Testing Results:

Our Heuristic evaluation consisted of two groups from class walking through our initial paper prototype design. After going through the two tasks with a group, we asked for each group's severity rating according to Neilsen's Heuristics and what event most commonly aligned with that. From these sessions, the heuristics violated were:

- Visibility of System Status: The system would not notify the user which way they were veering off of a path and how to correct it.
- User Control and Freedom: The application had no back button, no deletion of routes, or settings menu.
- Error Prevention: There was no way to cancel/stop a route, and they felt that more descriptions for which direction you are going would be helpful.
- Help users recognize, diagnose, and recover from errors: Users felt there were no notifications if something was wrong, and no way for help if something went wrong. Users also expressed a way to access contacts/emergency services while still being in the app.

From these heuristic evaluations, we received valuable feedback leading to our user testing. We refined each page to include a back button, and have more clarity in their functionalities. A deletion of routes, account creation, and settings pages were created. We also worked on more clear Screen Reader scripts, so we could effectively communicate the designs and functions to our next, visually impaired users.

For our first Usability Test, as previously mentioned, the Blind Services Coordinator (BSC) was our participant. During this interview, the BSC was first concerned that there were no indications of hazards, like crosswalks, to the user while traveling. This tester also expressed that they prefer when logins are linked to an existing account, so that they can use Face-ID to log in. Finally, the user liked that there was an option for multiple routes to be displayed, but that there was no way to tell the difference between routes when pulled up.

From this Usability Test 1, we updated our prototype to include an alert while traveling that a crosswalk with a stop sign is approaching, once the user was within

10 ft. of the cross walk. We also updated our login screens to have the option for SSO, Apple and Google login. Finally, we added a distance column to each route when it is displayed, so that the user has some indication of which route they will be going.

Due to scheduling, Usability Tests 2 and 3 were conducted in a series with no updates between user feedback. These interviews showed that while the stop sign hazard was useful, other hazards, like tabling in the MUC, are also something they want to be notified of. Users also felt that there were too many screens/steps to actually starting a route, and that that could frustrate some users. Finally, it was noted that a Settings menu and typing options would be helpful.

From these Usability Tests, we first added more hazard pop-up options, like the tabling in the MUC, to notify users of unexpected hazards that they may need to navigate. To address the issue of too many screens, we consolidated a few steps to one function, and now that process takes at most two screens to start a route. For our app, we also implemented the options to type wherever a search bar had been, instead of just voice-to-text. Finally, a settings page was created and an access button was added to the landing page.

Final Paper Prototype:

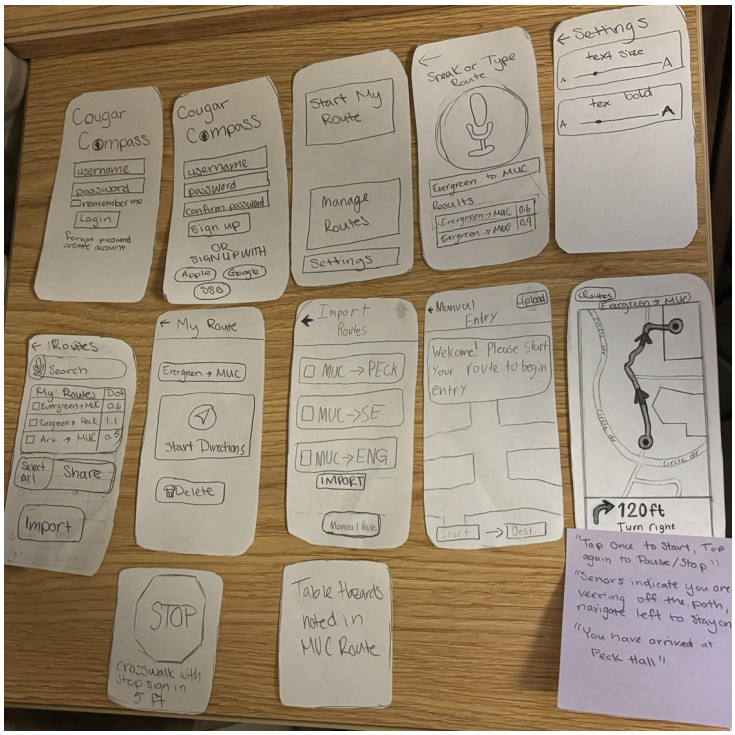
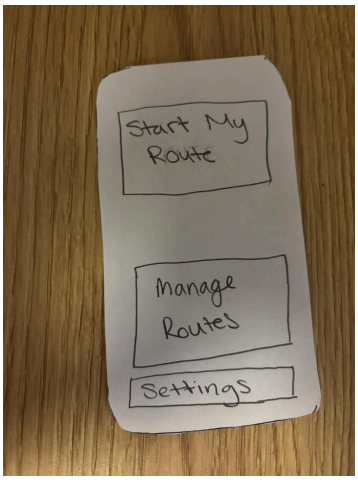


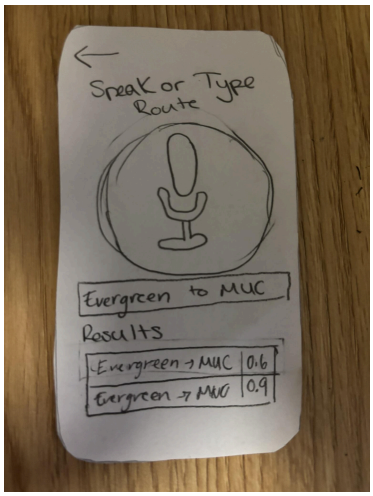
Figure 2: Final Paper Prototype (Sensors not included)

Our final paper prototype reflects changes made to improve user flow and customization, increase safety, and fix crucial mistakes. Changes from the initial prototype that occurred throughout heuristic evaluation and user testing to create the final paper prototype include adding a create account page, adding back buttons and delete options to increase user control, adding a settings page with the ability to increase text size and darkness, adding additional pop-ups, the combination of an initial task of alerting the user to crosswalks, and streamlining/combining a few screens to make starting a route and managing user routes simpler. Additionally, better alerts were added, and audio cues were modified. Simplicity was once again prioritized along with a focus on user control.

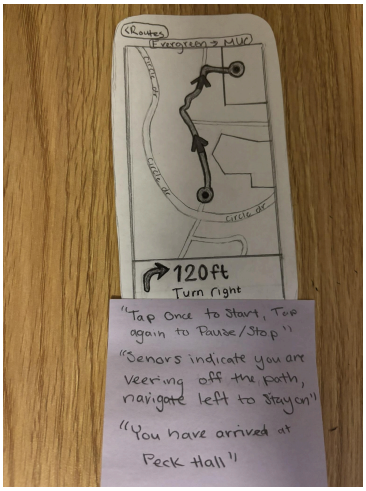
Task 1: Walking Independently Using Familiar Routes



A user has the option to “Start My Route”, which will take them to a page with a microphone for voice entry or a search bar.

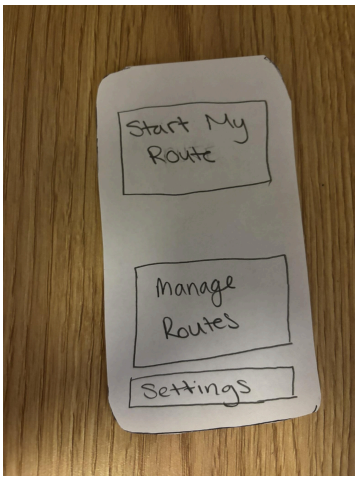


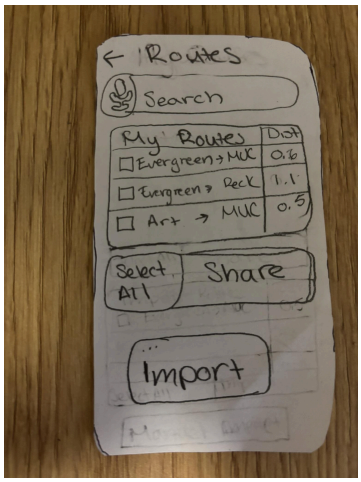
The user can tap to speak where they want to go or search it in the search box, with top results popping up below.

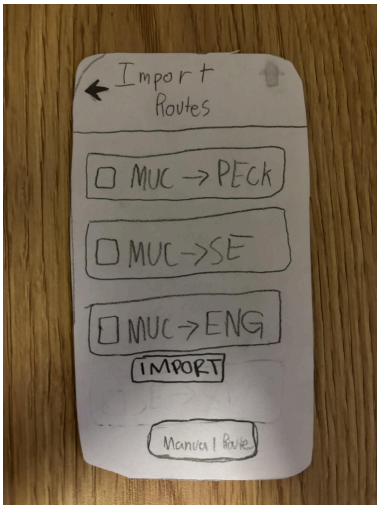


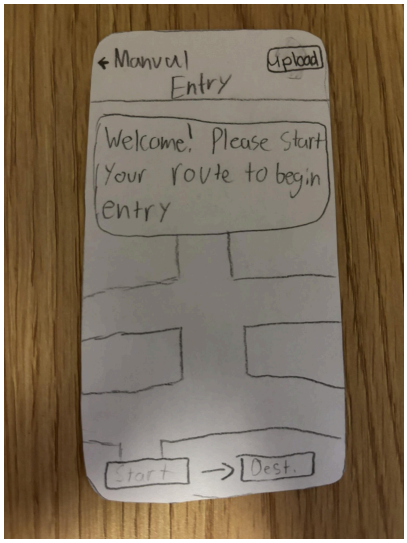
Users click on the route they want, and then listen to an audio cue of “Tap once to begin the route, tap again to pause.” The user will be given real-time audio cues about directions and warned if they are leaving the path. They will be told when they arrive and taken back to the home screen.

Task 2: Sharing Route “Memories” with Other Students

<table>
<tr>
<td> 

The user chooses to manage their routes, leading them to a screen listing all the routes currently saved to their account with the option to share.</td>
<td> 

The user clicks on all pre-existing routes in their database with the option to “select all” and then share. If a user wants to share a manually created, personal route that is not yet in their database, they must click “import.</td>
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<td> 

The user selects the “Manual Route” button</td>
<td> 

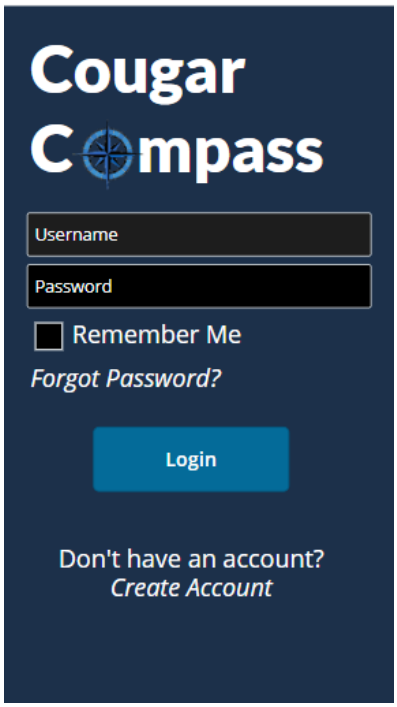
The user enters their start and end destination and begins walking, tapping “upload” when completed. The user is taken back to the home page and steps 1-3 are completed.</td>
</tr>
</table>

Digital Mockup Overview:

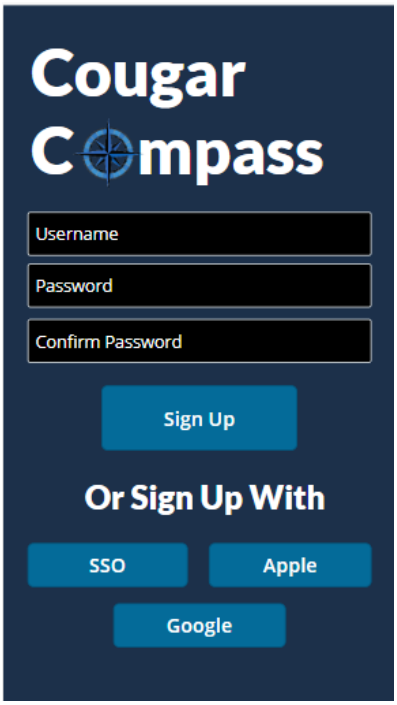


Figure 3: An overview image of Cougar Compass functionalities.

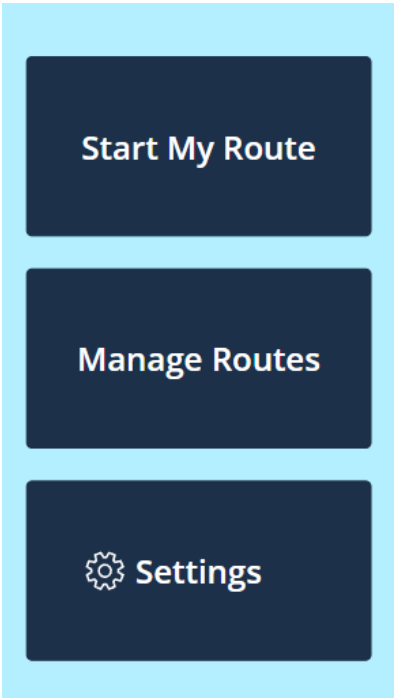
Digital Mockup: Task 1



A new user can create an account or sign up with their SIUE account (SSO), Apple Account, or Google Account Credentials.

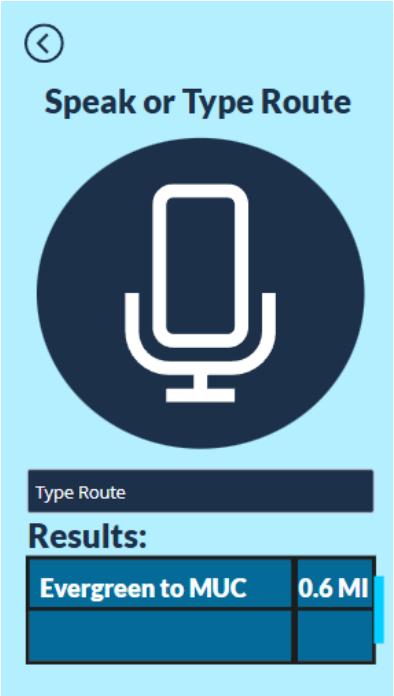


A user will log in to the application by providing a username and password. A “remember me” option is available, and users will not be automatically logged out each time they close the app.



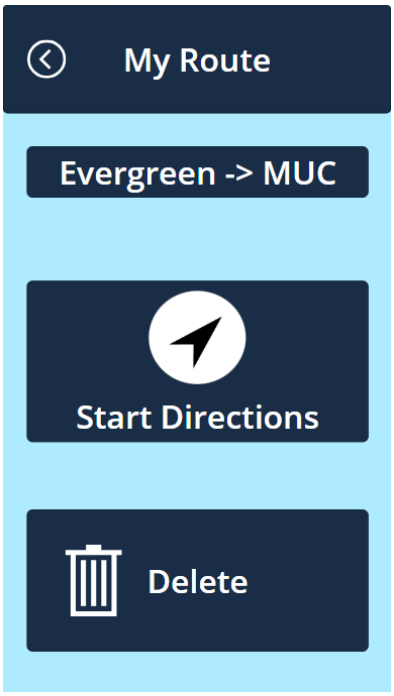
On the home page, the user can press “Start My Route” to be taken to the Search Page. If “Manage Routes” is chosen, the user would be taken to a list of their

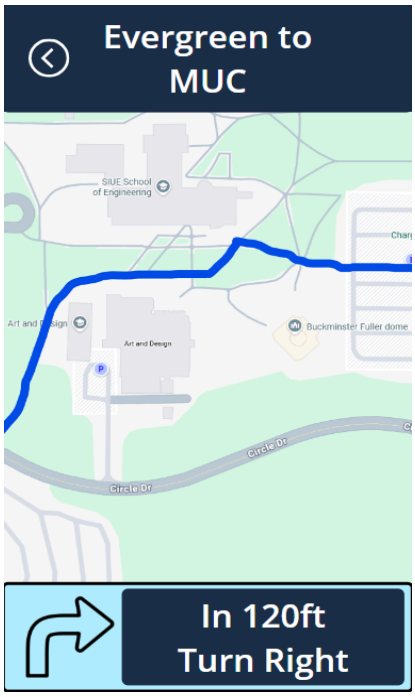
available routes. When one is tapped, they would be taken to the “My Route” page displayed below.



The user can either manually enter a route in the search bar or they can hold the microphone icon to use voice-to-text to search for a route. Top results appear in the Results scrollable table with their distances. A user taps on their desired route.

When a user has clicked on a route, they can “Start Directions” to begin auditory cues and directions as they are walking. They can also delete the route if, for example, it was a manually entered route and they decide they no longer want it.



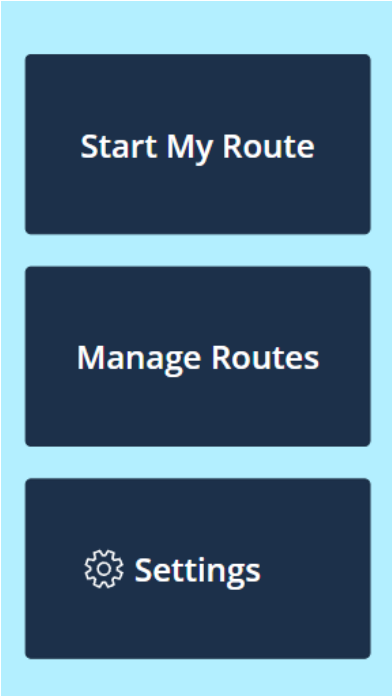


A map to the user's destination is pulled up. Sensors in the ground and inside buildings help provide GPS capabilities to interact with the app. When a student veers off the path, going outside of the sensor boundary, the app will tell the user they have left the path/route and what direction they should turn to get back onto it. As they approach turns, a user will be told which way to turn, and they will be alerted when approaching a crosswalk. When they have arrived, the app will tell them so and the user will be taken back to the home screen. Pressing the back arrow at the top can stop the route and take the user back to the description page of "My Route."

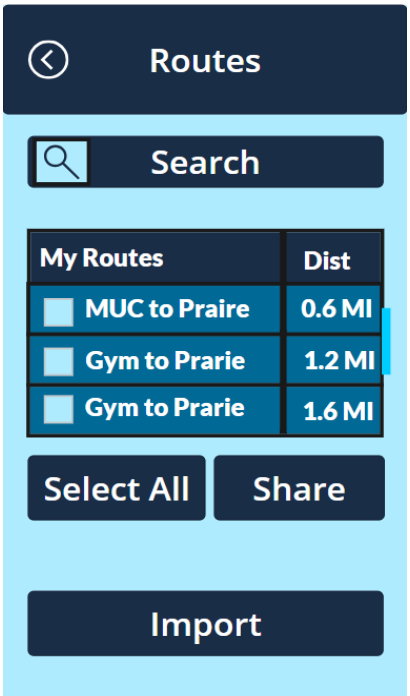
Example of a pop-up alert for a crosswalk nearby



Digital Mockup: Task 2



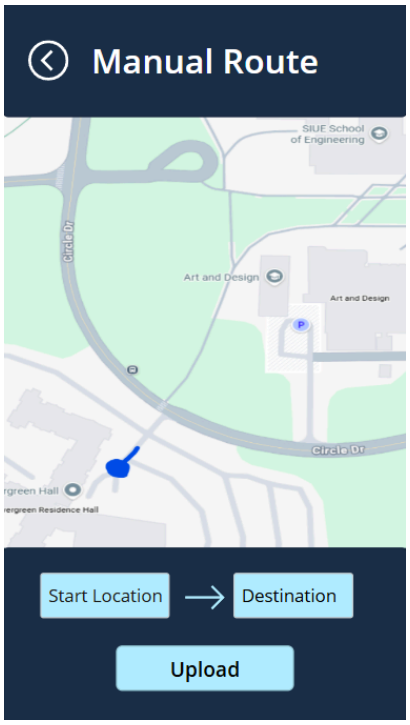
Assuming a user has already logged in, on the home page, they can select “Manage Routes.”



Under the “Manage Routes” page, a user can see a list of all current routes in their database. They can select them individually using the checkbox or “Select all.” Pressing “Share” will send the selected routes to the app-wide database where they can be imported by other users. If a user wants to create a manual route to share with another student, they must first press “Import.”

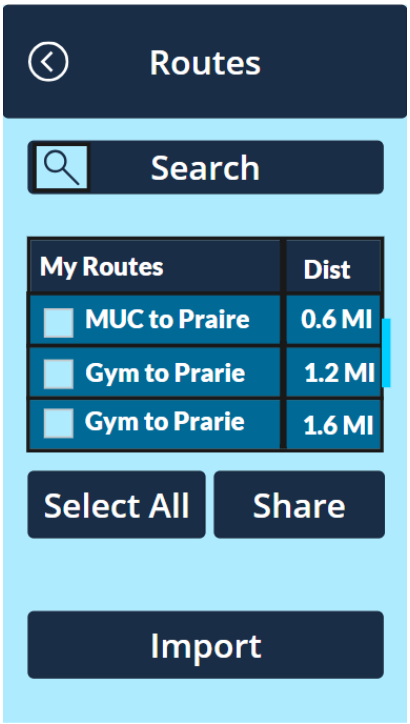


Under the “Import Routes” Page, users have the option to manually create a new route using “Manual Route.”



A user enters their starting and ending location with a prompt from the screen reader to tap the screen once to start and tap again to stop. The user, likely with a seeing individual acting as an aide, will then begin to walk the route, pressing “Upload” when they have completed it. This route will now be saved in their “My Routes” database.

With the new, manually created route saved in the user’s database, they can select it and click “Share.”



Transition from Paper to Digital

As we made the change from a physical paper prototype to a digital mockup, we had some changes to the overall design beyond just adding color. One such change was making buttons a more uniform size. An example occurs on our homepage, which originally included two large buttons with the settings as a small bar at the bottom. Now all buttons are the same size. Another such change is moving buttons to be in more convenient areas for the user. One such case is the manual entry, which used to have the upload in the top right corner. In our digital mockup, we moved it to be closer to the start - stop area. We also enlarged some text. On the map screen, the text saying where the user was going to and from used to be smaller, but with the digital prototype we gave it more room and a larger font, which can help with reading. We also added language options and a log out option to the settings page. We used a very small selection of color in our app (only 4 total, outside of the map itself), and made sure that there were no situations where we would have dark text on a dark background or light text on a light background.This color scheme achieves level AA compliance with the Web Content Accessibility Compliance.

Changes Based on Critique

In response to feedback, we simplified the overall navigation flow by reducing the number of steps required to start a route and consolidating features into a more cohesive home screen. We also improved accessibility by adding features such as adjustable text size, clearer labels, and support for both voice and manual input. Enhancements to system feedback—such as clearer instructions, hazard alerts, and off-route guidance—were implemented to address user concerns about clarity and safety. These changes resulted in a more intuitive, efficient, and accessible design.

Discussion:

Over the course of the iterative design process, we faced many functional issues that were not thought of during the initial design phase. During our initial design, we thought we were focusing on the absolute needs for our app to function, while being user-friendly. However, the heuristic design showed us that we had no way to

back out of a process and the app was not really “free.” The next testing also helped us improve the flow of the app in general. Overall, we have learned that the iterative design process helps identify flaws that us, as designers, have overlooked, and continue improvements as feedback comes in. Many small changes add up to a big difference that greatly improves the product.

This process let us gradually make changes to our design in a way that was not jarring to us or our product testers. As our changes were made, users were able to have some, if not most, of the same feel and layout to our app, just with the improvements requested. This helped us get user feedback in a way that also was not overwhelming to adjust for designs.

Overall, our tasks have not changed from usability testing. The tasks we made were relatively straightforward from the perspective of our participants, so this helped us to focus on the design, rather than needing to adjust tasks. Ultimately, we did end up incorporating a task that was not chosen as a main task into the route navigation, because it was requested by multiple users.

We believe that we could have used more iterations of our design, but were limited due to course time and participant availability. Ideally, to start our project, we would have included more than one walking interview and conducted a focus group for what solutions students would have wanted on campus. Due to time, we could not do this. We also believe more usability tests with Screen Reader functionality would have been very helpful for feedback and independent navigation, but again, we were limited by time constraints.

Appendix

Heuristic Evaluation Notes:

Evaluation from Group 2:

Names: Yakun, Anthony, Bryan, Cole

Visibility of system status – 1 (which direction are you veering off?)

Match between system and the real world - 0
User control and freedom – 1 (adding a route is a bit too free. What if you did it on accident? Can I delete a route?)
Consistency and standards - 0
Error prevention – 3 (Cancel a route, stop a route, which way are you going)
Recognition rather than recall - 0
Flexibility and efficiency of use – 0 (the purpose of a few buttons was confusing but deemed good overall)
Aesthetic and minimalist design – 0
Help users recognize, diagnose, and recover from errors - 2 (same thing as before: don't let the user know when something is wrong)
Help and documentation - 1 (on the side button -- reach out to emergency services)

Notes:
-- No create account screen, oops; just sign in
-- Make sure to have the screen reader say "Click screen once to start, click again to stop"
-- Which way am I veering off in between the sensors? (include audio cues)
-- crosswalk safety (one of our 6 tasks not implemented in this feature, but really quite important to get someone safely to their destination)
-- No indicator to pause or exit route
-- Confusion for the upload arrow (is this a back button?)

Evaluation from Group 4:

Names: Alexander, Nathan, Jalen, Sean

Visibility of system status - 3 (going off the path)
Match between system and the real world - 0
User control and freedom – 1 (said it was straightforward) add settings
Consistency and standards – 0
Error prevention – 2 (deleting a route, object in path)
Recognition rather than recall – 0 (concise screen reading, good. Tap once is good)
Flexibility and efficiency of use – 0
Aesthetic and minimalist design – 0 (liked there was no clutter)
Help users recognize, diagnose, and recover from errors - 1 (same as error prevention; not something for an option to call police, or help; include non – emergency; a quick way to exit the app or access contacts without exiting map)

Help and documentation – 0

Notes:
-- “You are off the path.” Well, how do I get back on? (ex. “Go right.”
-- The upload button was understood well in this group
-- Import meaning/purpose needs clarity when someone hasn’t used it before
-- They liked the modulation (each thing only goes one layer deep then goes back).
-- Add settings: change language option

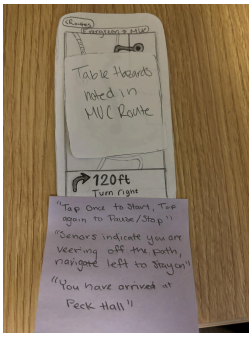
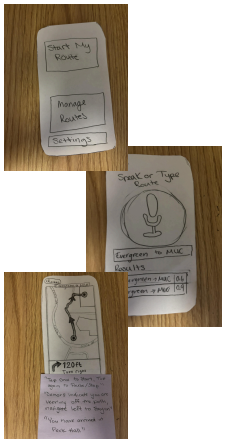
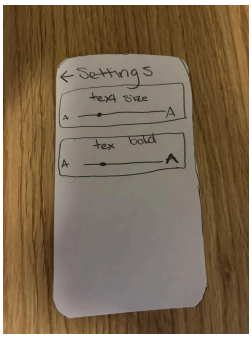
Usability Test 1 Results

Original Image	Issue Identification	Severity	Revision	Revision Image
	**No Indication of Hazards** Did not alert the user to hazards such as crosswalks **Heuristic:** Error Prevention	4	App now will alert when within a distance from a crosswalk	
	**Minimal Login Options** There was a way for users to create native accounts, but not third-party sign ons. The user also would have to relogin each	2	Added third-party login options, like apple, google, and SIUE SSO. Added a remember me option	

	time they use the app **Heuristic:** Flexibility and Efficiency of Use			
	**Minimal Sign Up Options** Extension from login, no way to sign up using third party apps **Heuristic:** Flexibility and Efficiency of Use	2	Added third-party sign up options	
	**Multiple Routes** The app would show routes available, but not always give a route the user wants **Heuristic:** Consistency and Standards	1	Listing distances with how long each route is to differentiate between one another.	

Usability Test 2/3

Usability tests 2 and 3 were conducted one after another due to last minute scheduling needs. So, all updates were only reflected after Test 3.

Original Image	Issue Identification	Severity	Revision	Revision Image
	**No Indication of Hazards** Did not alert the user to hazards such as pop up tables or wet floors **Heuristic:** Error Prevention	4	App now will alert when within a distance from a crosswalk, and other user reported hazards	
Too many screens to photograph	**Too Many Screens** Users stated there were too many screens to start route **Heuristic:** Flexibility and Efficiency of Use	2	App now only has 2 screens at most to start route	
No previous screen	**No Settings Menu** Users wanted ability to adjust button and text size	1	Users can now control text size which will automaticall y adjust button size	

	Heuristic: Visibility of System Status			

Contributions:

Emmorie Brown: 50% Testing Results, Digital Mockup, Discussion
Faith Smith: 50% Problem and Solution Overview, Testing Process, Initial Paper Prototype, Final Paper Prototype, and Appendix